Fatemeh SHIRVANI

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Education

Institut Polytechnique de Paris **September 2025 – Expected September 2027**
Master of Science
Interaction, Graphics & Design
Selected coursework: Advanced Computer Graphics; Image Synthesis; Computer Animation; Digital Representation and Analysis of Shapes (Discrete Differential Geometry); Image Analysis & Computer Vision; Human-Computer Interaction for Mixed Reality.

Isfahan University of Technology **September 2020 – June 2024**
Bachelor of Science
Computer Engineering, Intelligent Systems (GPA 16.16/20)

Experience

R&D Intern, 3D Graphics & VR – EssilorLuxottica **June – August 2026**
Supervised by Thomas Charbonneau.
- Reconstructed an existing indoor refraction room from photographs as a navigable VR scene using 3D Gaussian Splatting.
- Developed the capture-to-reconstruction workflow with COLMAP and assessed novel-view quality, identifying fidelity issues around reflective, low-texture, and thin surfaces.
- Tested VR rendering and interaction in Unity with VRSplat and OpenXR, comparing visual quality, navigation, and runtime behavior to inform the deployment approach.

Research Assistant, Interactive Systems – DIVA Group **October 2025 – June 2026**
Institut Polytechnique de Paris. Supervised by Prof. James Eagan. **Manuscript in preparation.**
- Designed and implemented coordinated Sorter and Outliner instruments for polymorphic structural editing, applicable across different kinds of structured content.
- Built a shared node model and representation-specific adapters in JavaScript and Webstrates, reusing navigation and reordering operations while preserving each artifact's constraints.
- Demonstrated the approach on documents, slide decks, and photo galleries, examining consistency between structural edits, source content, and rendered views.

Research Assistant, Motor Learning – University of Tehran **June – September 2024**
Supervised by Dr. Fariba Bahrami.
- Studied motor learning as participants practiced the new task of drawing shapes with their foot on a tablet.
- Extracted features from recorded trajectories using Python and MATLAB, including drawing accuracy, movement time, and smoothness, to characterize changes with practice.
- Explored learning patterns with clustering, then compared support-vector classifiers and random forests to distinguish early from later stages, reaching 81% accuracy in leave-one-subject-out testing.

Selected Projects

Variance Soft Shadow Mapping (March – April 2026)
- Implemented Yang et al.'s VSSM (2010) in WebGPU/WGSL with summed-area moment queries for blocker-depth estimation and soft-shadow filtering.
- Extended the method with adaptive quadtree subdivision and exact PCF fallback in unstable regions, reducing light leaking and contact-shadow artifacts on thin foliage.

Plant Growth Modeling (February 2026)
- Generated a growing 3D tree in C++/OpenGL using stochastic L-system branching and staged branch and leaf activation.
- Integrated smooth seasonal transitions, hierarchical wind motion, leaf detachment, and alpha-tested foliage shadows to keep animation and rendering temporally coherent.

Simulation & Geometry Processing (January – February 2026)
- Loop subdivision: adjacency-based refinement of triangular meshes in C++/OpenGL.
- Rigid-body dynamics: box and cylinder contacts with impulse response, restitution, and Coulomb friction.
- SPH fluids: 2D solver with grid-based neighbor search and pressure and viscosity forces.

Teaching Assistant

- Digital System Design (Winter 2023)
- Advanced Programming (Winter 2022)
- Discrete Mathematical Structures (Winter 2022)
- Data Structures (Fall 2022)

Skills

- Languages: C++, C#, Python, JavaScript, MATLAB
- Graphics/3D: OpenGL, WebGPU, GLSL, WGSL, Unity, Blender, 3DGS
- ML/CV tools: PyTorch, scikit-learn, OpenCV, NumPy

Awards & Activities

- **IP Paris INNOV'NIGHT 2026** – winning team.
- **Volunteer work** – Graphics Staff, AICUP (IUT, 2021 and 2022); Technical Staff, REACT (SUT, 2023).

Languages

- English (Fluent, TOEFL: 107/120)
- French (B1)
- Persian (Native)

Interests

Climbing; setar (Iranian instrument); philosophy.